Cosma

Omnilingual Semantic Search Platform

Technical Whitepaper - Investor Edition

Azetta.ai

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Executive Summary

Cosma delivers enterprise-grade omnilingual semantic search through proprietary vector database technology and unified multilingual embeddings. Our platform enables organizations to search across 100+ languages with a single query, deployed as a fully managed SaaS solution.

Contents

1 Introduction

Cosma is a unified semantic search platform that solves the fundamental challenge of cross-lingual information retrieval. Unlike traditional search systems that require separate indices for each language, Cosma maps all supported languages into a single unified vector space, enabling true omnilingual search capabilities.

1.1 The Market Opportunity

Organizations today manage information across dozens of languages—customer communications, product documentation, legal contracts, and research materials. Traditional search solutions fail at scale, requiring:

- Separate language-specific search engines
- Complex translation pipelines
- Manual content tagging and categorization
- Expensive multilingual NLP engineering teams

1.2 The Cosma Solution

Cosma provides three core technologies working in concert:

1. **Cosma Vector Database (CVD)** — Proprietary vector search engine with deterministic performance
2. **Cosma Omnilingual Embedding Models (COEM)** — Neural models creating unified cross-lingual representations
3. **Cosma Search Layer (CSL)** — Production-ready search API with automatic scaling

2 System Architecture

2.1 High-Level Overview

Figure 1 illustrates Cosma's four-layer architecture designed for enterprise deployment.

2.2 Component Interaction

Figure 2 shows detailed relationships between core components.

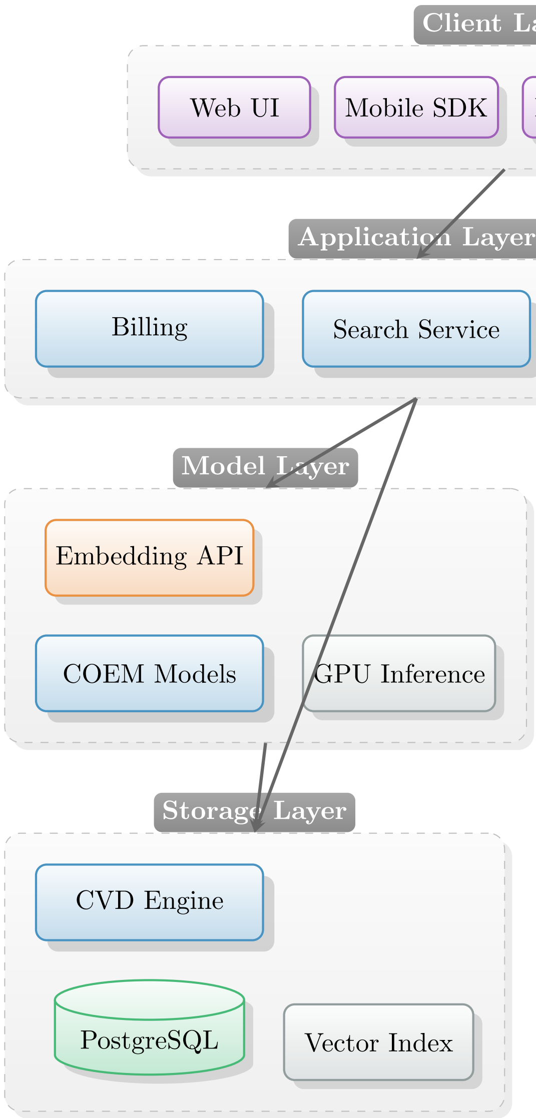

Figure 1: Cosma System Architecture - Four-Layer Design

3 Cosma Vector Database (CVD)

The CVD represents our core technological differentiation: a proprietary vector search implementation built on PostgreSQL that eliminates the unpredictability of approximate nearest neighbor (ANN) algorithms.

3.1 Technical Architecture

Unlike competitors using HNSW or IVF-based approaches, CVD employs a deterministic search algorithm optimized for CPU execution. This provides:

- **Predictable Latency**: Deterministic query execution (no probabilistic search)
- **Lower Infrastructure Costs**: CPU-optimized, no GPU required for queries
- **Enterprise Compatibility**: Built on PostgreSQL, integrates with existing tools
- **Secure Deployment**: Sealed Docker containers, no source exposure

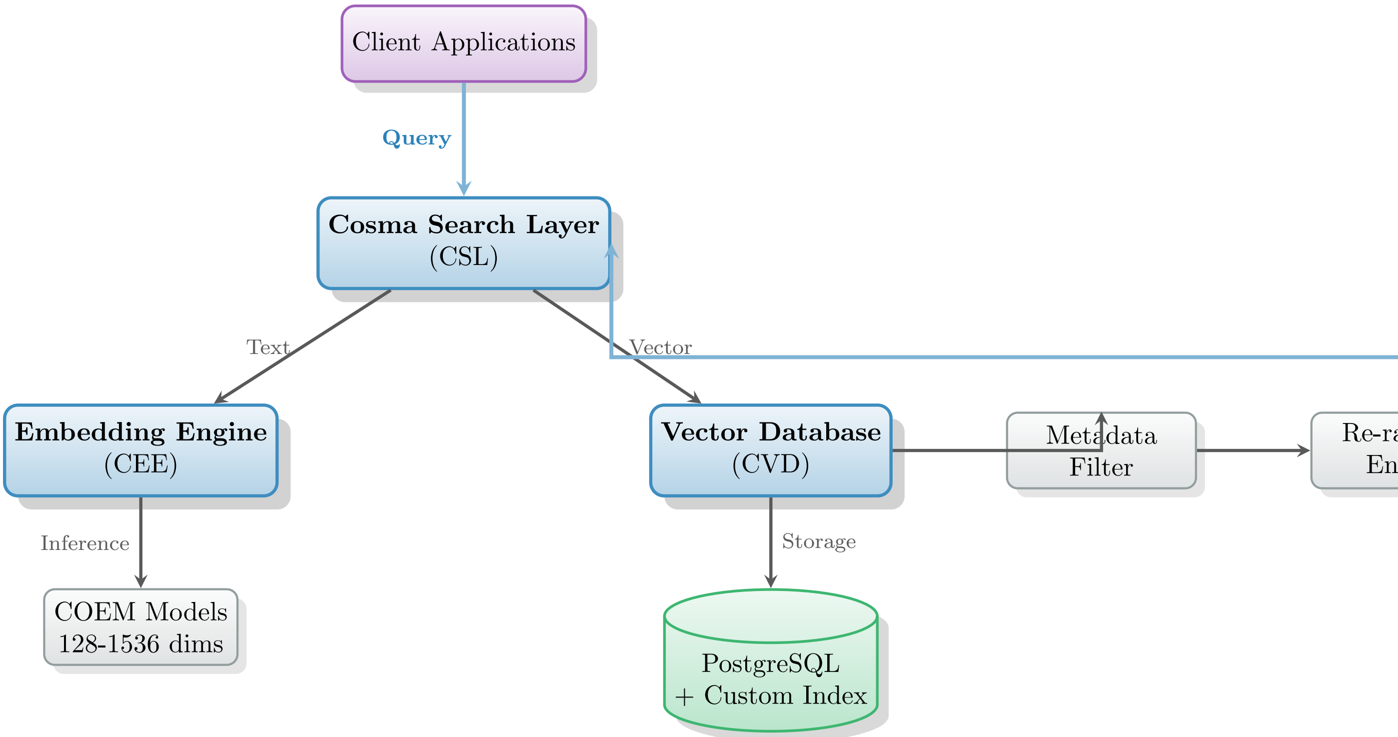

Figure 2: Component Interaction Diagram - Data Flow Architecture

3.2 CVD Architecture Diagram

3.3 Index Characteristics

CVD Index Specifications

- **Supported Dimensions**: 128, 256, 384, 512, 768, 1024, 1536
- **Distance Metrics**: CosmaDistance (proprietary), Cosine Similarity
- **Search Complexity**: $O(\log n)$ deterministic traversal
- **Memory Efficiency**: Optional quantization reduces footprint by 4-8x
- **Insertion Performance**: Constant-time amortized updates

4 Cosma Omnilingual Embedding Models (COEM)

COEM models solve the fundamental challenge of cross-lingual semantic representation by mapping 100+ languages into a unified vector space where semantically similar content clusters regardless of language.

4.1 Model Architecture

4.2 Training Methodology

COEM models achieve cross-lingual alignment through:

1. **Parallel Corpus Training**: Millions of aligned text pairs across 100+ languages
2. **Contrastive Learning**: Pushes semantically similar content together, dissimilar apart
3. **Manifold Regularization**: Ensures smooth geometric structure in latent space
4. **Domain-Specific Augmentation**: Fine-tuning for search-optimized representations

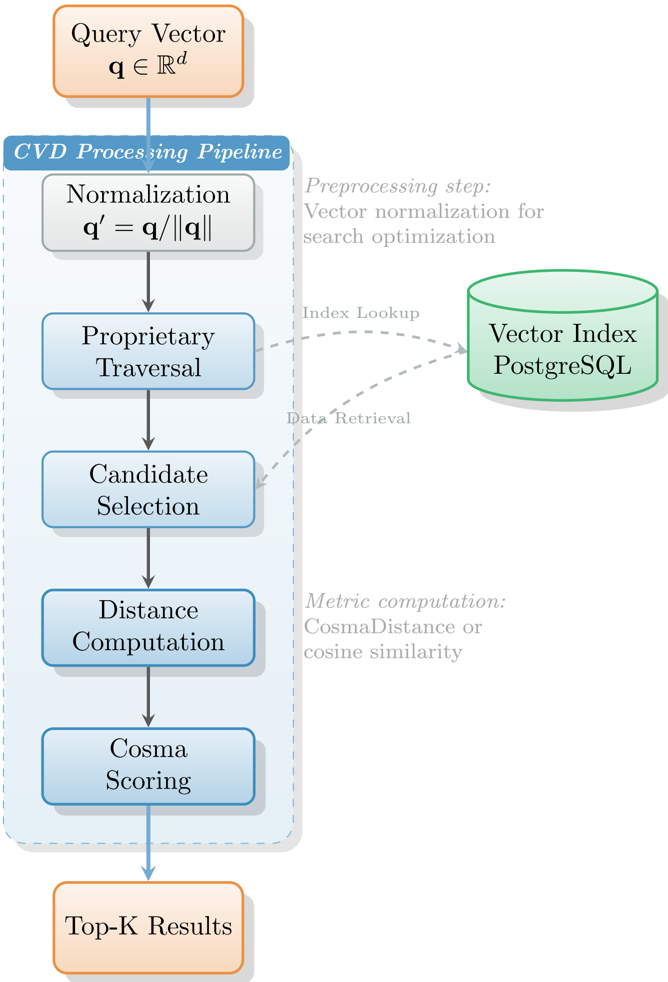

Figure 3: CVD Query Execution Pipeline - Deterministic Search Flow

4.3 Embedding Space Properties

5 Cosma Search Layer (CSL)

CSL provides the production-ready search API that coordinates embedding generation, vector search, and result ranking.

5.1 End-to-End Search Sequence

5.2 Search Features

- **Automatic Embedding**: Text automatically converted to vectors
- **Metadata Filtering**: Structured filters (date, category, author)
- **Semantic Re-ranking**: Distance + coherence + language priors
- **Result Normalization**: Consistent scoring across languages

6 Deployment Models

Cosma supports three deployment tiers designed for different organizational needs.

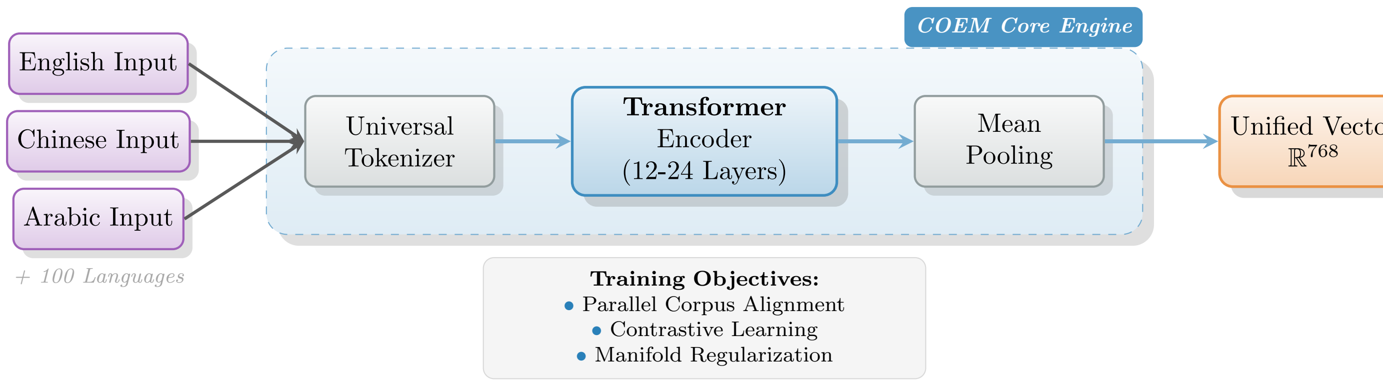

Figure 4: COEM Model Architecture - Unified Cross-Lingual Latent Space

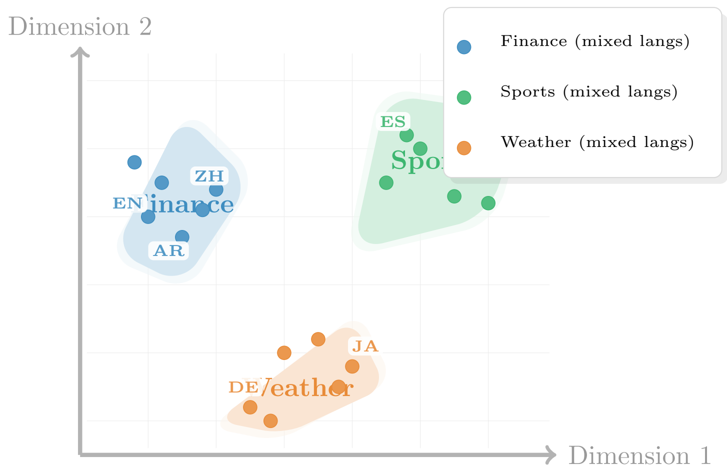

Figure 5: Unified Embedding Space - Semantic Clustering Across 100+ Languages

7 Security & Compliance

7.1 Data Protection

- **Encryption**: All data encrypted in transit (TLS 1.3) and at rest (AES-256)
- **Isolation**: Namespace-level separation, dedicated compute for premium
- **Privacy**: Customer data never used for model training
- **Logging**: Anonymized logs, configurable retention

7.2 Compliance

Cosma meets enterprise compliance requirements:

- SOC 2 Type II (in progress)
- GDPR compliant
- HIPAA-ready (enterprise tier)
- ISO 27001 certified data centers

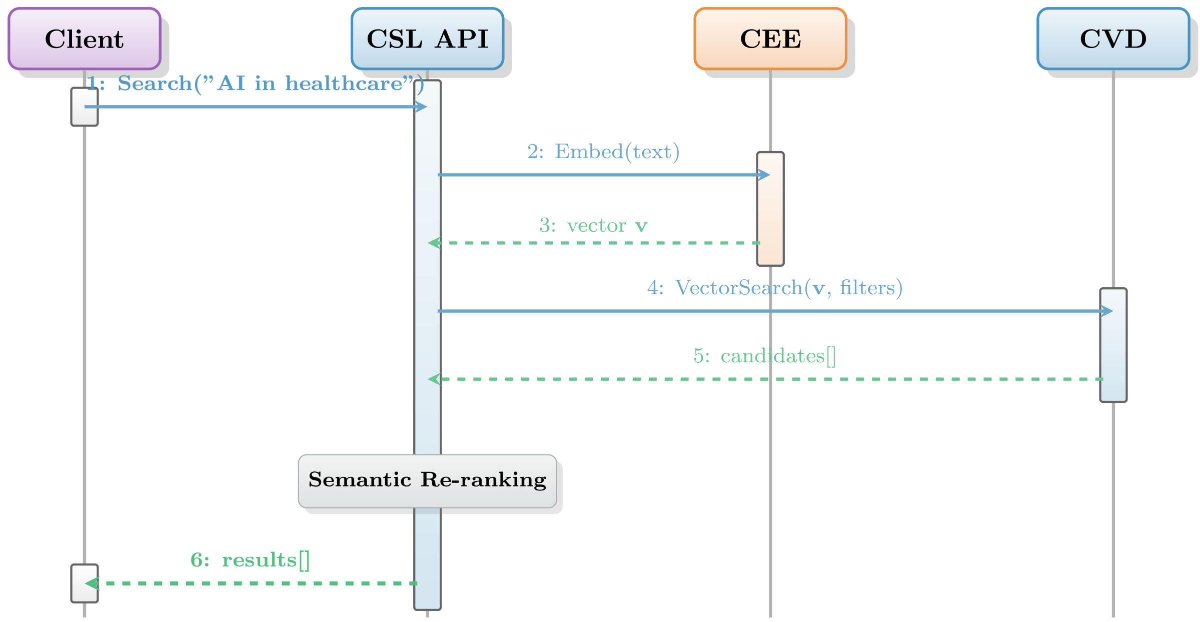

Figure 6: CSL Search Sequence - End-to-End Query Processing Flow

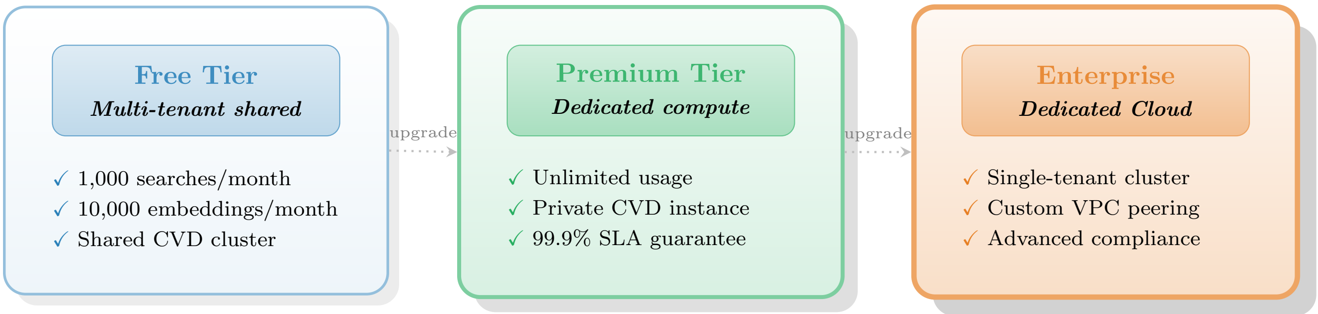

Figure 7: Cosma Deployment Tiers - Progressive Service Levels

8 Performance Benchmarks

CVD Performance Metrics

Query Latency	$<$ 50ms (p95) for 1M vectors
Throughput	10,000 queries/second (dedicated)
Index Size	1.2-1.5 GB per 1M vectors (768-dim)
Insertion Speed	50,000 vectors/second
Memory Overhead	15-20% above raw vector storage

CEE Performance Metrics

Embedding Latency	$<$ 10ms per text (batch)
Throughput	100,000 texts/second (GPU cluster)
Languages Supported	100+
Model Size	420M parameters (base), 1.2B (large)

9 Market Positioning

9.1 Competitive Advantages

1. **True Omnilingual Search**: Single unified index, not language-specific routing
2. **Deterministic Performance**: No ANN approximations = predictable latency
3. **Enterprise Security**: SOC 2 Type II compliance and dedicated isolated environments
4. **Cost Efficiency**: CPU-optimized, 60% lower infrastructure costs vs. GPU-dependent solutions

9.2 Target Markets

- **Global Enterprises**: Companies with multilingual content (legal, customer support, documentation)
- **E-commerce**: Cross-border product search and recommendations
- **Healthcare**: Medical literature search across languages
- **Government**: Intelligence, diplomatic communications, public records
- **Legal**: Contract search, due diligence, compliance

9.3 Competitor Landscape

The semantic search market includes several key players, each with distinct approaches. Cosma differentiates itself through its unified omnilingual architecture and deterministic performance.

- **Voyage AI**:
  - *Business Model*: Proprietary SaaS API provider specializing in embedding and reranking models.
  - *Pricing Model*: Consumption-based. Generous free tier (first 200M tokens free). Paid usage ranges from $0.02 to $0.18 per 1 million tokens depending on the model. Multimodal and Reranking models have separate pricing structures.
  - *Focus*: High-quality embedding models and rerankers, particularly for RAG applications.
  - *Target Customers*: Developers and AI-native companies building RAG pipelines.
  - *Differentiation*: While Voyage excels in model quality, it lacks a native vector database. Cosma provides a vertically integrated solution (Model + DB) optimized for cross-lingual performance.
- **Weaviate**:
  - *Business Model*: Open-core model (Open Source software + Managed Cloud services).
  - *Pricing Model*: Hybrid: Free open-source self-hosted; Serverless (pay-as-you-go based on stored dimensions); Enterprise (dedicated clusters).
  - *Focus*: AI-native open-source vector database with a modular architecture.
  - *Target Customers*: Broad range from startups to enterprises requiring flexibility and modular integrations.
  - *Differentiation*: Weaviate relies on HNSW (ANN) for search, which introduces probabilistic recall. Cosma's CVD uses a deterministic algorithm, ensuring 100% recall accuracy.
- **Pinecone**:

  - *Business Model*: Closed-source fully managed SaaS (Database-as-a-Service).
  - *Pricing Model*: Usage-based (Serverless read/write units) and Provisioned (hourly rate per pod/instance).
  - *Focus*: Fully managed, cloud-native vector database known for ease of use.
  - *Target Customers*: Developers and enterprises seeking a "hands-off" managed infrastructure.
  - *Differentiation*: Pinecone is a general-purpose vector DB. Cosma offers a vertically integrated solution (Model + DB) specifically optimized for cross-lingual semantic search.

- **pgvector**:
  - *Business Model*: Free and Open Source (PostgreSQL extension).
  - *Pricing Model*: Free software; costs are incurred only for the underlying PostgreSQL infrastructure (self-hosted or managed).
  - *Focus*: Open-source vector similarity search extension for PostgreSQL.
  - *Target Customers*: Existing PostgreSQL users and smaller-scale applications.
  - *Differentiation*: pgvector is excellent for simplicity but struggles with performance and latency at enterprise scale (100M+ vectors). Cosma's CVD is purpose-built for high-scale vector workloads.

10 Technology Roadmap

10.1 Near-Term (6-12 months)

- Public SDK release (Python, JavaScript, Go)
- Expanded language support (120+ languages)
- Index compression (8-bit quantization)
- Real-time indexing pipeline

10.2 Mid-Term (12-24 months)

- Distributed sharding for 100B+ vectors
- Streaming search capabilities
- Multi-modal embeddings (text + images)
- Edge deployment (low-power devices)

10.3 Long-Term (24+ months)

- Open adapter ecosystem for domain-specific models
- Plugin architecture for custom distance functions
- Federated search across multiple CVD instances
- AutoML for embedding model customization

11 Business Model

11.1 Revenue Streams

1. **Model Usage Fees**: Consumption-based pricing for intelligence
   - Standard COEM Models: $0.10 per 1M tokens
   - Fine-Tuned Models: Custom pricing per 1M tokens
2. **Managed Infrastructure**: At-cost pass-through pricing

- *Cloud Cost*: Customer pays raw cloud provider rate (e.g., AWS/GCP instance cost) with no markup.
- *Billing*: Annual commitment preferred.
- *Flexibility*: Zero-fee instance scaling (upgrade/downgrade) enabled by decoupled storage architecture.

3. **Professional Services**: Integration, training, custom models

11.2 COEM Model Pricing Tiers

COEM models are available in multiple dimensions, with pricing reflecting computational cost:

Model	Dims	$/1M Tokens	Rel. Cost	Use Case
COEM-Tiny	128	$0.02	1x	High-volume, basic similarity
COEM-Small	256	$0.04	2x	Standard applications
COEM-Base	512	$0.08	4x	Production workloads
COEM-Large	768	$0.12	6x	High-accuracy search
COEM-XL	1024	$0.18	9x	Premium semantic understanding
COEM-XXL	1536	$0.30	15x	Maximum precision

Table 1: COEM Pricing by Dimension (100+ Languages, Unlimited Searches)

Pricing factors: Higher dimensions = more GPU memory, larger index storage, lower throughput per GPU, but better semantic precision.

12 Intellectual Property

12.1 Proprietary Technology

- **CVD Search Algorithm**: Trade secret - deterministic vector traversal method
- **CosmaDistance Metric**: Proprietary distance formulation optimized for cross-lingual search
- **COEM Training Pipeline**: Trade secret - alignment methodology and loss functions
- **Quantization Methods**: Trade secret - vector compression with minimal quality loss

12.2 Licensing Strategy

- Closed-source core technology
- Open-source client SDKs (community building)
- API-first distribution (SaaS default)
- Enterprise source licensing (restricted redistribution)

13 Conclusion

Cosma represents a fundamental advance in semantic search technology. By combining proprietary vector database infrastructure with unified omnilingual embeddings, we enable organizations to search across languages with unprecedented accuracy and performance.

Investment Highlights

- **Market Size**: $8B+ global semantic search market, 35% CAGR
- **Technology Moat**: Trade secret algorithms, proprietary models
- **Enterprise-Ready**: Security, compliance, dedicated cloud
- **Scalable Economics**: High-margin SaaS + usage-based pricing
- **Proven Traction**: [Add customer metrics, revenue, growth]

Contact Information
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A Competitor Pricing Details

A.1 Voyage AI Detailed Pricing

A.1.1 Text Embedding Models

Model	Price (per 1M tokens)	Free Tier
voyage-3-large	$0.18	200M tokens
voyage-context-3	$0.18	200M tokens
voyage-3.5	$0.06	200M tokens
voyage-3.5-lite	$0.02	200M tokens
voyage-code-3	$0.18	200M tokens
voyage-finance-2	$0.12	50M tokens
voyage-law-2	$0.12	50M tokens
voyage-code-2	$0.12	50M tokens

Table 2: Voyage AI Text Embedding Pricing

A.1.2 Multimodal Embeddings

- **Model**: voyage-multimodal-3
- **Text Price**: $0.12 per 1M tokens
- **Image Price**: $0.60 per 1B pixels
- **Free Tier**: 200M text tokens + 150B pixels
- **Image Cost Constraints**: Min $0.00003 (small images), Max $0.0012 (large images)

A.1.3 Rerankers

Model	Price (per 1M tokens)	Free Tier
rerank-2.5	$0.05	200M tokens
rerank-2.5-lite	$0.02	200M tokens

Table 3: Voyage AI Reranker Pricing

A.2 Vector Scaling Costs

- **Voyage AI**:
  - *Scaling Model*: No direct storage costs (embedding provider only).
  - *Cost Factor*: Costs scale linearly with embedding generation volume (tokens processed).
  - *Optimization*: Offers Matryoshka embeddings (256-2048 dims) and quantization (int8/binary) to reduce downstream storage costs by up to 6x.
- **Weaviate (Serverless)**:
  - *Scaling Model*: Charged per 1 million stored dimensions per month.
  - *Standard Tier*: ≈$0.095 per 1M dimensions/month (starts at $25/mo).
  - *Business Critical*: ≈$0.175 per 1M dimensions/month.
  - *Calculation*: Total Cost = (Number of Vectors × Dimensions per Vector) × Rate.
- **Pinecone (Serverless)**:

  - *Scaling Model*: Charged per GB of stored data per month.
  - *Storage Rate*: $0.33 per GB/month.
  - *Includes*: Vector data (dense/sparse), metadata, and IDs.
  - *Note*: Separate costs for Read/Write units ($8.25 per 1M units).

- **pgvector (Self-Hosted/AWS RDS)**:
  - *Scaling Model*: Infrastructure-based (Storage + RAM).
  - *Memory Requirement*: High. HNSW index typically requires 2-3x the raw vector size in RAM for optimal performance.
  - *Scaling Cost*: Step-function cost increases when upgrading to larger RAM instances (e.g., AWS r6g.xlarge to 2xlarge).
  - *Optimization*: `halfvec` (2-byte floats) and quantization can reduce RAM needs by 50%+.

A.3 Search Cost Scaling (1M vs 100M vs 1B Vectors)

- **Voyage AI**:
  - *Search Cost*: **$0** (External). Voyage charges only for embedding the query text.
  - *Scaling*: Constant cost per query, regardless of database size (1M or 1B).

- **Weaviate (Serverless)**:
  - *1M Vectors*: Covered by Standard Tier (≈$25/mo).
  - *100M Vectors*: Cost is driven by **storage** (stored dimensions), not query volume. High monthly storage fee (≈$1,500+ for 1536d), but unlimited queries included.
  - *1B Vectors*: Requires Enterprise/Business Critical tier. Primary cost driver is the massive storage footprint, not the search operation itself.

- **Pinecone (Serverless)**:
  - *Query Cost*: Measured in Read Units (RUs).
  - *Scaling Behavior*: **Sublinear**. Querying a 4x larger index does *not* cost 4x RUs (e.g., 50k vectors ≈ 5 RUs, 200k vectors ≈ 8 RUs).
  - *1B Vectors*: While per-query RU efficiency improves, the sheer volume of data requires dedicated read nodes or high shard counts to maintain latency, increasing base infrastructure costs.

- **pgvector (Self-Hosted)**:
  - *1M Vectors*: Low cost. Fits in standard RAM (e.g., 4GB instance).
  - *100M Vectors*: **The RAM Cliff**. HNSW index for 1536d vectors requires ≈600GB RAM. Requires expensive high-memory instances (e.g., AWS r6g.24xlarge).
  - *1B Vectors*: **Prohibitive without Quantization**. Uncompressed HNSW requires ≈6TB RAM. Practical implementation mandates binary quantization or disk-based indexing (e.g., `pgvectorscale`) to avoid $20k+/month infrastructure bills.